SeqGPT: A Constrained Transformer Agent for the Inverse Design of Multi-Panel Composite Structures

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Résumé

La conception inverse de structures composites exige de réconcilier des cibles de performance continues avec un espace de design discret et contraint. C'est une approche classique lorsque des stratégies d'optimisation numérique bi-niveaux sont employées. Ce problème inverse combinatoire devient particulièrement critique dans les structures multi-panneaux en raison du blending—une contrainte de continuité ou compatibilité globale entre les différents empilements. Ce travail introduit SeqGPT, un agent Transformer conditionnel conçu pour remplacer les méthodes itératives, coûteuses en temps de calcul. Afin de garantir la continuité globale et la validité manufacturière par construction, nous employons une stratégie de décodage neuro-symbolique hybride. À chaque pli, le modèle prédit une distribution conditionnelle qui guide une Constrained Beam Search, dans laquelle toute branche violant les règles de blending est strictement élaguée. Les expériences numériques sur le benchmark du fer à cheval à 18 panneaux démontrent que SeqGPT génère des solutions quasi-instantanément avec des performances en flambement comparables aux méthodes évolutionnaires, offrant une accélération considérable par rapport à l'état de l'art.

Mots-clés

IA Neuro-symbolique, Transformer, Modèle conditionnel, Optimisation, Composites.

Abstract

Optimizing composite stacking sequences to match continuous targets (e.g., Lamination or Buckling Parameters) with discrete manufacturing constraints represents a challenging combinatorial inverse problem that regularly occurs in composite design especially when numerical optimization approaches are used (bi-step, bi-level configurations). In multi-panel configurations, this complexity is further intensified by blending—a global compatibility/continuity requirement between the different panel stackings. This study presents SeqGPT, a conditional Transformer agent developed to replace computationally expensive iterative methods. To ensure both global continuity and manufacturing feasibility by construction, we implemented a hybrid neuro-symbolic decoding strategy. SeqGPT predicts a conditional distribution that guides a Constrained Beam Search, where any branch violating blending rules is strictly pruned. Numerical experiments on the 18-panel horseshoe benchmark demonstrate that SeqGPT generates solutions near-instantaneously with buckling performance comparable to evolutionary methods, offering a significant speed-up compared to the state of the art.

Keywords

Neuro-Symbolic AI, Transformer, Conditional Model, Optimization, Composite Structures, Constrained Generation.

1 Introduction

The contemporary aerospace industry is undergoing a progressive substitution of traditional metal alloys by Carbon Fiber Reinforced Polymer (CFRP) composites—which now account for approximately 53% of the Airbus A350 XWB structural weight. This transition constitutes a complete redefinition of structural design. Unlike isotropic metals, composite structures rely on the superposition of unidirectional plies to resist multi-axial forces, allowing for elastic tailoring where fiber orientations are aligned with principal stress directions. Nevertheless, physically manufacturing such components requires a specific discrete stacking sequence. The design space is typically restricted to a discrete set of orientations $\{0°, 90°, \pm 45°\}$, creating a non-convex optimization problem with exponential complexity. Furthermore, solutions must satisfy "hard constraints"—such as symmetry, balance, and contiguity—which act as mandatory design rules [2]. Matching this theoretical potential with manufacturing constraints remains a significant combinatorial challenge. This complexity is further amplified when scaling up to large "one-shot" structures divided into multiple adjacent panels because of the global topological dependencies inherent to large-scale aerospace structures, where the sequence of a panel is strictly conditioned by the geometric continuity (blending) with its neighbors. This introduces the requirement for blending, turning the optimization into a coupled problem where decisions on one panel propagate constraints to its neighbors, often leading to prohibitive convergence times [6].

State-of-the-art methods generally adopt a two-step optimization strategy: first optimizing continuous structural pa-

rameters (e.g., lamination or buckling parameters), and then retrieving a discrete stacking sequence that matches these targets [1]. In contrast, Irisarri et al. [2] proposed a monolithic formulation in which blending constraints are enforced directly through the discrete optimization of Stacking Sequence Tables (SST) using a Genetic Algorithm. While effective, this approach suffers from a strong combinatorial growth of the design space. To restore computational efficiency while guaranteeing blending, the Search Propagation Direction (SPD) method was introduced [6], enforcing continuity by construction through iterative ply drops or insertions. However, the retrieval of admissible stacking sequences within this framework still relies on iterative stochastic solvers, which remains a bottleneck for real-time interactive design.

Recently, deep generative models have emerged as a promising alternative to stochastic search. Notable examples include the Physics-Informed CGAN by Qiu et al. [5] and LayupFormer introduced by Xie et al. [7]. While these state-of-the-art methods successfully map mechanical targets to valid stacking sequences, they operate at the single-laminate level, producing one layup for a given structure without considering inter-laminate blending constraints. These models are trained for a specific optimization problem, and extending them to new configurations typically requires costly surrogate retraining.

In this work, we share the same intuition as [7], reformulating stacking sequence retrieval as a conditional generation task. Building on the general solver paradigm enabled by the bi-step optimization strategy and the Search Propagation Direction (SPD) method [6], we developed SeqGPT, a Transformer-based agent trained to satisfy manufacturing rules by conditioning predictions on non-dimensional buckling parameters. While SPD ensures computational efficiency and blending compliance at a global scale, our hybrid neuro-symbolic approach, inspired by NeuroLogic Decoding [4], enforces strict global continuity. In this framework, SeqGPT provides probabilistic guidance conditioned on the target structural parameters, while a coupled Constrained Beam Search strictly prunes invalid branches to satisfy blending rules by construction.

The remainder of this paper is organized as follows. Section 2 details the state-of-the-art methods. Section 3 describes the SeqGPT architecture and the neuro-symbolic decoding logic. Finally, Section 4 evaluates the performance of the proposed method against traditional stochastic solvers.

2 State of the Art: From Composite Optimization to Generative Inference

2.1 The Bi-Step Strategy: Bridging Global Sizing and Local Retrieval

State-of-the-art methods generally adopt a two-step strategy (Fig. 1) to handle the combinatorial complexity of composite design [1].

The first step idealizes the entire multi-panel structure as a continuous domain. At this stage, the solver evaluates the complete structural assembly—incorporating global geometry, boundary conditions, and internal load redistributions—to minimize the total mass. Instead of discrete plies, the design variables are relaxed into continuous quantities [1]. Specifically, the solver identifies a global optimum vector $\boldsymbol{\xi}^*$ for each individual panel. This target vector comprises four non-dimensional Nemeth parameters $(\alpha, \beta, \gamma, \delta)$ that describe the local physics of buckling, alongside a fifth parameter, N, representing the target continuous thickness (or optimal number of plies). Since this relaxed feasible domain is convex, gradient-based algorithms can efficiently process the global Finite Element Analysis (FEA) to provide these continuous mechanical targets [1].

The second step, **Stacking Sequence Retrieval (SSR)**, constitutes the main computational bottleneck. At this stage, the structure must be translated back into discrete ply angles $\theta \in \Theta = \{0°, \pm 45°, 90°\}$. In the standard bi-step approach, the SSR is formulated as a purely local inverse problem for each panel:

$$\hat{S} = \arg\min_{S \in \mathcal{X}} \|\Phi(S) - \boldsymbol{\xi}^*\|^2, \tag{1}$$

where $\mathcal{X} \subset \Theta^N$ is defined as the discrete design space of **valid** stacking sequences of length N. Specifically, a sequence belongs to $\mathcal{X}$ if and only if it satisfies a representative subset of critical manufacturing constraints (MCs), selected from the comprehensive guidelines by Irisarri et al. [2].

Table 1: Selected subset of critical Manufacturing Constraints (MCs) adapted from [2], defining the feasible design space $\mathcal{X}$.

Scope	ID	Requirement and Technical Description
Local	**MC1**	**Symmetry:** Stacking sequences must be mirrored with respect to the mid-plane, ensuring that the coupling matrix B is zero and decoupling in-plane and bending responses.
	MC2	**Contiguity:** Limits the number of consecutive plies with identical orientation (typically ≤ 4) to mitigate the risk of matrix cracking and large-scale delamination.
	MC3	**Balance:** Requires an equal number of $+45°$ and $-45°$ plies within the laminate to eliminate shear–extension coupling and ensure membrane stability.
	MC4	**Disorientation:** Limits the orientation change between adjacent plies ($0°/90°$ is not allowed) to reduce interlaminar shear stresses and improve fatigue life.
	MC5	**Proportion:** Requires each principal ply direction ($0°, 90°, \pm 45°$) to constitute between a minimum (typically 10%) and a maximum (typically 66%) of the total laminate to guarantee sufficient multi-axial strength and damage tolerance.
Global	**MC0**	**Blending:** Each ply present in a thinner panel must continue into adjacent thicker panels to ensure a continuous load path across the structure.

As detailed in Table 1, these defining rules are partitioned into two categories: **local constraints** (MC1–MC5), which

ensure the structural integrity of each isolated laminate, and a **global constraint** (MC0), which enforces ply continuity (blending) between adjacent panels. While the first step efficiently resolves the global mechanics, the SSR step introduces a severe combinatorial challenge. Since Φ is non-injective (i.e., multiple stacking sequences can produce the same continuous targets), this creates a highly non-convex search landscape filled with local optima where standard solvers struggle to converge [6]. Crucially, formulating the SSR as a set of independent local problems strictly confines the search space to the local rules (MC1–MC5). By definition, this panel-by-panel approach cannot enforce the inter-laminate global compatibility constraint (MC0).

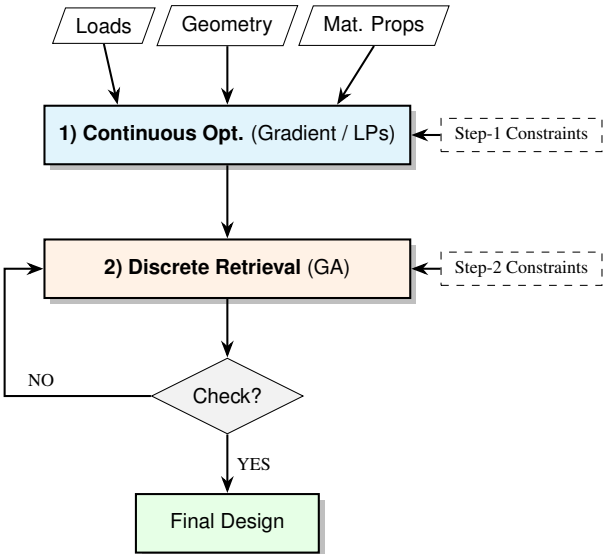

Figure 1: The standard Bi-Step Optimization workflow. Step 1 uses fast gradients on the global continuous assembly. Step 2 retrieves the discrete sequences locally, a process that suffers from combinatorial complexity and fails to guarantee global blending.

2.2 Enforcing Global Constraints: Blending and the SPD Strategy

To overcome this structural incompatibility and enforce the global blending constraint (MC0, Fig. 2), the optimization must account for the strict topological dependencies between adjacent panels. Rather than solving isolated problems, the stacking sequence of each laminate becomes tightly conditioned by its neighbors. As highlighted by Scardaoni and Montemurro, this non-trivial coupling transforms the sequence retrieval phase into a massively coupled Constraint Satisfaction Problem (CSP) that must be solved at the scale of the entire structure [6].

To address this global coupling, Irisarri et al. [2] proposed the **Stacking Sequence Table (SST)**. This method optimizes a single master sequence from which all panels are derived by truncation. While this guarantees blending by definition, it forces the optimizer—typically a Genetic Algorithm—to evaluate the entire table against the whole structure at every iteration, leading to prohibitive computational costs.

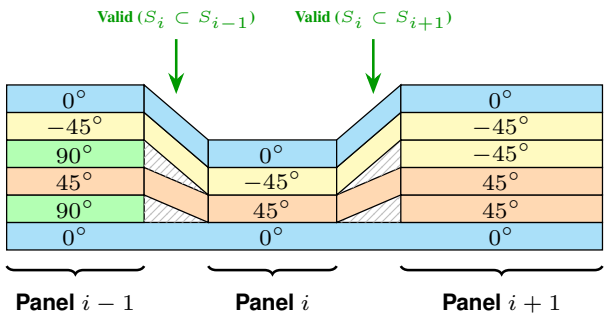

Figure 2: Cross-sectional view of the Blending Constraint. The intermediate panel S_i acts as a topological bridge ($S_i \subset S_{i-1}$ and $S_i \subset S_{i+1}$), ensuring global continuity between neighbors S_{i-1} and S_{i+1} that may otherwise possess mutually incompatible sequences.

To reduce the global complexity, we adopt the *Search Propagation Direction* (SPD) strategy proposed by Scardaoni and Montemurro [6]. SPD decomposes the structure into a directed path of local problems (Fig. 3). In this framework, the stacking sequences of adjacent panels S_p and S_q—comprising N_p and N_q plies respectively—are coupled via Propagation Maps $\pi_{p\to q}$ to ensure structural continuity.

- **A ply drop** ($N_p > N_q$) uses the map $\pi_{p\to q}$ to select N_q plies to retain from S_p while preserving their relative order.
- **A ply insertion** ($N_p < N_q$) ensures that all plies from S_p are inherited, and additional plies are inserted into S_q via a strictly increasing injective map of indices.

By fixing this propagation order, the global blending constraint is enforced *by construction*: once the parent sequence S_p is defined, all child sequences S_q along the path are automatically valid. The remaining task reduces to assigning the free variables (ply orientations) at the root and along the propagation path.

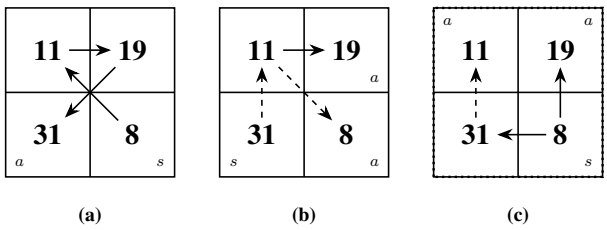

Figure 3: Illustration of sequence propagation mechanisms. The numbers within the grids indicate the sequence length N. Initial sequences are marked with s and end sequences with a. Solid arrows ($\rightarrow$) denote ply insertion, while dashed arrows ($\dashrightarrow$) denote ply drop. (a) Valid propagation utilizing only insertions. (b) Valid propagation mixing insertion and drop with a different start sequence. (c) Invalid cyclic propagation.

In cases (a) and (b), the Search Propagation Direction (SPD) [6] ensures global blending by construction, as the generation paths systematically enforce compatibility rules across the physical boundaries of adjacent panels.

In case (c), however, the propagation tree branches out, leaving physically adjacent panels (such as the 11-ply and 19-ply laminates) on separate generation branches. Because there is no direct parent-to-child relationship explicitly evaluated across their shared boundary, the generative model does not enforce blending rules between them. As highlighted by Scardaoni and Montemurro [6], such a propagation strategy is invalid because it provides no mathematical guarantees that these independently generated sequences will be compatible.

2.3 From Stochastic Search to Generative Inference

Although the SPD method simplifies the global topology into a sequence of local problems, the instantiation of local variables typically relies on iterative stochastic solvers. To overcome this computational latency, recent works have proposed replacing the heuristic search with Deep Generative Models, treating stacking sequence retrieval as a structured generation task.

Conditional GANs (CGAN) were proposed by Qiu et al. [5] as a Physics-Informed approach to generate sequences from mechanical targets. However, as noted by Xie et al. [7], their application is typically limited to restricted laminate families and their stochastic nature lacks interpretability, obscuring the physical relationship between requirements and layups. Furthermore, as confirmed in our own experiments, this adversarial approach suffers from inherent training instabilities and is prone to *mode collapse*, leading to a severe loss of design diversity when scaling to larger sequences.

Transformer-Based Architectures were concurrently investigated by Xie et al. [7], who introduced *LayupFormer* to independently validate the potential of these models for composite design. Formulated as an Encoder-Decoder sequence-to-sequence task. While sharing this linguistic analogy, their implementation relies on explicit grammar masking applied to the decoder outputs [7]. This forces the model to produce valid tokens (symmetry, balance) by zeroing out invalid probabilities after the Softmax layer. Additionally, their approach employs a complex hybrid loss function (Eq. 7 in [7]) combining cross-entropy with auxiliary physics penalties. Computing these penalties requires training and maintaining a high-fidelity differentiable surrogate (predictor). This introduces architectural overhead and requires tuning weighting hyperparameters to balance the competing objectives of syntactic validity and physical compliance.

Our proposed neuro-symbolic approach, in contrast, introduces a strategy that decouples physical performance from topological constraints. The neural agent, SeqGPT, approximates the inverse distribution $P(S \mid \boldsymbol{\xi}^*)$ to inherently satisfy local constraints (MC1–MC5) without intermediate surrogates. To address the global topology, our Constrained Beam Search practically implements the Search Propagation Direction (SPD) to enforce blending (MC0). By treating inter-panel derivation as a constrained tree search, symbolic logic masks strictly prune topologically invalid ply drops and insertions. Throughout this generative process, SeqGPT acts as a physics-driven heuristic to evaluate and rank the remaining valid branches, transforming the combinatorial complexity of multi-panel assemblies into a tractable, beam-limited exploration.

3 Methodology

The methodology is presented from three perspectives: (1) The definition of a domain-specific tokenization strategy that maps physical laminate properties to discrete sequence tokens. (2) The architecture of the SeqGPT agent (3) The neuro-symbolic decoding protocol

3.1 AI Modeling and Tokenization Strategy

As introduced in Section 2.1, the generative model is conditioned on the continuous target vector $\boldsymbol{\xi}^*$ and the target laminate thickness N. To integrate these physical targets into the neural architecture, $\boldsymbol{\xi}^*$ and N are concatenated and directly mapped into the model's latent space via a linear projection layer.

The Symmetry constraint (MC1) allows us to optimize the generation process: the neural agent generates only the first half of the stacking sequence, denoted by a length n. The parity of the total thickness N dictates the exact symmetry mapping:

- **For an even thickness** ($N = 2n$), the sequence is mirrored purely across the mid-plane.
- **For an odd thickness** ($N = 2n - 1$), the sequence is mirrored around the central mid-plane ply. In our tokenization, the last generated token before EOS corresponds to this exact mid-plane ply. For example, given a target thickness of $N = 39$, the model generates a sequence of length $n = 20$, where the 20th token acts as the axis of symmetry.

Next-token prediction. Since ply orientations belong to a discrete set, we map the standard angles $\{0°, 90°, 45°, -45°\}$ to integer tokens $\{1, 2, 3, 4\}$. To handle variable-length sequences within the fixed-length architecture of Transformers, we introduce two structural tokens: Token 0 (`[PAD]`) is used for padding during batched training, and Token 5 (`[EOS]`) represents the End-of-Sequence token. During training, the `[EOS]` token is included as the ground-truth target following the final ply of each sequence providing a clear signal of sequence termination. However, it is not used as a stopping criterion during inference. Because the target thickness N is provided in the prompt, the required half-sequence length n is known in advance, and generation stops once n plies have been produced.

Initialization Prompt. To ensure outer surface protection (Damage Tolerance), we enforce a fixed boundary condition where all generated laminates must begin with a

$[45°, -45°]$ block, mapped to tokens $[3, 4]$. In our Decoder-Only architecture, the autoregressive generation is triggered by an initial context acting as a prompt and denoted as `[BOS]`. This initial state, `[BOS]`, encapsulates the physical conditioning vector $[PHYS] = (\boldsymbol{\xi}^*, N)$ alongside these fixed boundary tokens. This initializes the sequence as `[BOS]` $= [PHYS, 3, 4]$, ensuring that every generated layup is physically conditioned from the first step and satisfies the mandatory surface requirements by construction.

3.2 Architecture and Input Conditioning

We employ a conditional Transformer architecture based on the **minGPT** framework [3] (see Figure 4). The model is a lightweight Transformer Decoder composed of **3 layers** with **3 attention heads**. The embedding dimension is set to $d_{model} = 48$, resulting in approximately 83k trainable parameters. Following the standard minGPT architecture, each transformer block incorporates Layer Normalization, Masked Multi-Head Attention, and a Multilayer Perceptron (MLP) featuring a GELU non-linear activation function. Dropout layers are also included to act as a regularization technique and prevent overfitting.

We have two different input types: conditional input parameters $[PHYS] = (\boldsymbol{\xi}^*, N)$ and discrete ply tokens $\{1, 2, 3, 4\}$. To mix them, we project them into a shared space of dimension d_{model}.

Because Transformers require fixed-length input vectors while optimal laminates exhibit variable thicknesses, we standardize the sequence capacity based on our dataset's maximum thickness, $N_{\max} = 41$. Consequently, we define a maximum context window of $n_{\max} = \lceil N_{\max}/2 \rceil = 21$. During inference, SeqGPT autoregressively generates the half-sequence until n ply tokens have been produced. Any remaining sequence length up to $n_{\max}$ is subsequently filled with `[PAD]` tokens.

Feature Fusion and Positional Encoding.We use two projection layers to enter the common latent space:

- The **conditional input** vector of physical targets is processed by a Linear Layer. This projects floating-point values into vectors $v_{phys} \in \mathbb{R}^{d_{model}}$.
- The **discrete tokens** pass through a standard trainable Embedding Layer. This maps each integer to a dense vector $v_{ply} \in \mathbb{R}^{d_{model}}$.

We concatenate these vectors to form the input sequence, with the physical parameters prepended as a conditioning prompt for generation. Because transformers treat tokens as unordered, this architecture adds learned positional embeddings to the sequence vectors. This allows the model to capture the exact positional context of each token within the stack.

Attention Mechanism and Physical Alignment. During the self-attention calculation, every new token generated at step t attends back to all previous positions $\theta_{<t}$. Crucially, it also attends to the initial positions corresponding to v_{phys}. The attention weights implicitly learn to correlate specific ply orientations to match the physical target.

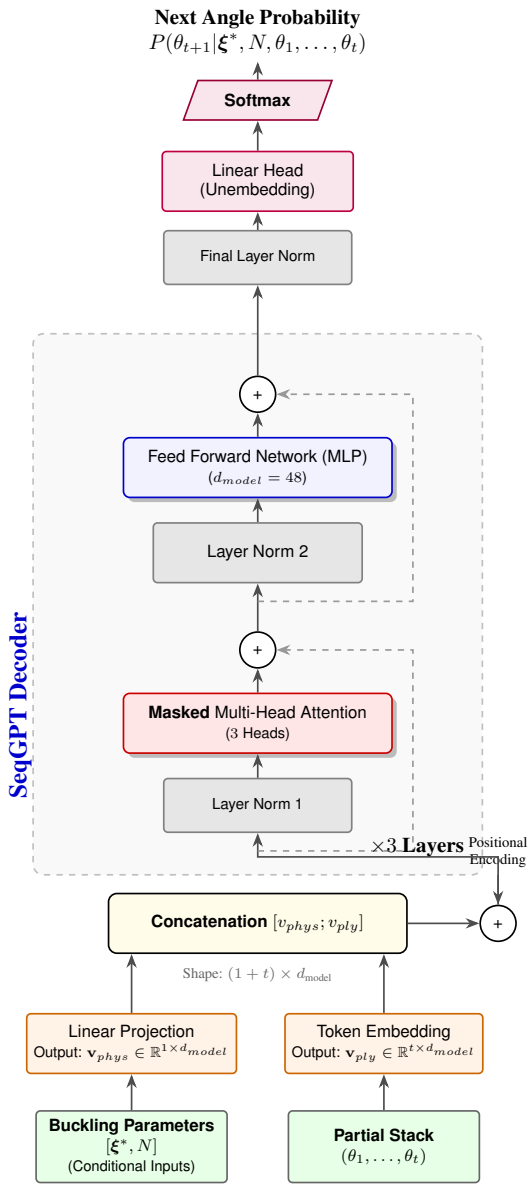

Figure 4: Architecture of the SeqGPT agent. The model is a lightweight Decoder-Only Transformer.

Training Objective and Masking. The model operates autoregressively: at each step it predicts the probability distribution of the next token conditioned on the sequence history and the physical context. The network is trained by minimizing the Cross-Entropy loss. However, the conditional input token `[PHYS]` and the padding token `[0]` must be excluded from the optimization. To achieve this, a binary mask m_t is introduced such that $m_t = 0$ for positions corresponding to physical parameters or padding tokens, and $m_t = 1$ for valid ply tokens. For a sequence with maximum length $n_{\max}$, the masked Cross-Entropy loss over the vocabulary space V is defined as

$$\mathcal{L} = -\sum_{t=1}^{n_{\max}} m_t \sum_{j=1}^{|V|} \hat{y}_{j,t} \log(y_{j,t}), \tag{2}$$

where $|V|$ denotes the vocabulary size, $\hat{y}_{j,t} \in \{0, 1\}$ is the one-hot encoded ground-truth token for class j at position t, and

$$y_{j,t} = P(\theta_t = j \mid \theta_{<t}, \boldsymbol{\xi}^*, N)$$

is the probability predicted by the softmax layer for token j at step t, conditioned on the previous plies and the target physical parameters. The objective optimized during training corresponds to the average of this loss over all tokens in a mini-batch.

3.3 Multi-Panel Generation via Constrained Decoding

SeqGPT efficiently generates isolated laminates that satisfy all local rules (MC1–MC5). However, multi-panel blending (MC0) requires deriving a child sequence S_c from a parent S_p. While SeqGPT's autoregressive nature intrinsically respects Markovian constraints like contiguity (MC2) and disorientation (MC4), it lacks visibility over the parent's available ply inventory. Consequently, locally optimal predictions can render the sequence infeasible later—for instance, dropping the parent's only 0° ply irreversibly violates the child's proportion constraint (MC5).

To prevent such conflicts and guarantee that the neural predictions align with both the blending topology (MC0) and the cumulative manufacturing rules (MC3, MC5), a child laminate is considered *blendable* with respect to its parent if and only if it satisfies the set of **hard blending rules** defined in Table 2. These rules (C1–C3) act as explicit symbolic masks within the decoding loop. Multi-panel generation therefore amounts to searching, in the discrete space of orientation sequences, for paths that remain valid under these combined constraints.

This induces a constrained tree search where each node corresponds to a partial stacking sequence and each edge to the addition of one ply. At each depth t, if a candidate ply violates any of the hard constraints (either by breaking the parent's topology or exhausting a critical orientation), the entire branch is pruned.

Figure 5 illustrates the constrained beam search procedure with a beam width of $K = 3$. At each step, the decoder retains the top-K ranked valid branches, transforming the exponential search into a tractable beam-limited exploration guided by SeqGPT.

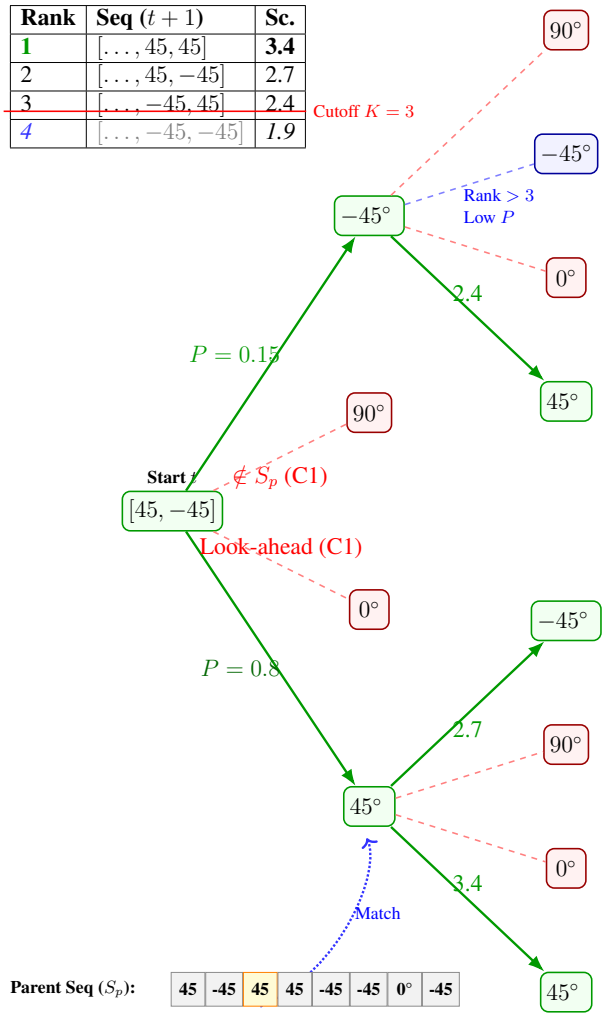

Figure 5: Neuro-Symbolic Constrained Beam Search for laminate blending. Candidate plies proposed by SeqGPT are filtered by hard blending rules with respect to the parent sequence S_p, then ranked using a probability–penalty score. Only the top-K valid branches are retained at each step.

SeqGPT acts as a probabilistic guide to rank the remaining valid candidates. Given a partial sequence $\theta_{<t}$, the score of a candidate ply is

$$\text{Score}(\theta_{t+1}) = \log P(\theta_{t+1} \mid \theta_{<t}, \boldsymbol{\xi}^*, N), \tag{3}$$

which favors mechanically plausible continuations conditioned on the optimal continuous parameters $\boldsymbol{\xi}^*$ and the target thickness N.

Table 2: Decoding Constraints for Blending. Hard constraints (C) strictly prune the search space to ensure manufacturing feasibility.

ID	Constraint	Description / Condition
C1	Sequence Heritage	**Drop:** Inherit plies from S_p sequentially ($S_c \subset S_p$). The total number of skipped parent plies must not exceed $N_p - N_c$. **Insertion:** Inherit all plies from S_p sequentially ($S_p \subset S_c$). The total number of newly generated plies must not exceed $N_c - N_p$.
C2	Balance	S_p must provide counter-angles to offset current S_c imbalance.
C3	Proportion	Sufficient angles of each orientation remain to satisfy the ply proportion constraint.

Optional soft constraint guidance. Beyond hard pruning, additional manufacturing heuristics can be incorporated as soft penalties that bias the ranking without affecting feasibility. In this case, the score becomes

$$\text{Score}_{penalty}(\theta_{t+1}) = \text{Score}(\theta_{t+1}) - \lambda \cdot \text{SC}(\theta_{t+1}),$$

where $\text{SC}(\theta_{t+1}) \geq 0$ denotes a penalty term measuring the violation of a soft manufacturing preference, and λ controls its influence.
Specifically, in the "drop" mode, we can define a continuity penalty SC that favors subsequences that maintain geometric homogeneity. Let k_t and k_{t+1} be the indices in the parent sequence $\mathcal{S}_p$ of the plies selected for the child sequence at steps t and $t+1$. The jump distance is defined as $j = k_{t+1} - (k_t + 1)$. The penalty is then proportional to j, favoring the selection of contiguous plies ($j = 0$) and penalizing the omission of parent material ($j > 0$).
Hard constraints (C1–C3) remain strictly enforced through pruning, while soft penalties only affect the ordering of valid candidates within the beam.
To guarantee robustness, a **Conservative Fallback Strategy** is introduced. If the constrained beam search fails to produce a valid sequence for a target thickness N_{target} (often due to overly restrictive constraints), the decoder is rerun with an increased thickness $N_{\text{target}} + 1$. This process is repeated until a feasible laminate is obtained. In the worst case, when $N_{\text{target}} = N_{\text{parent}}$, the parent stacking sequence is copied. This ensures that a manufacturable solution is always returned.

4 Numerical Results and Discussion

This section presents the evaluation of the proposed framework, moving from intrinsic model assessment to industrial validation. Section 4.1 first characterizes the generative performance independently of the beam search solver, evaluating the model's capacity to strictly respect manufacturing constraints, accurately target buckling parameters, and provide interpretability through self-attention mechanisms. Section 4.2 then validates the full neuro-symbolic framework on the standard 18-panel horseshoe benchmark. Finally, Section 4.3 discusses these results by comparison to the baseline in terms of mechanical margins and computational efficiency.

4.1 Single Panel Generation with SeqGPT

4.1.1 Diversity–Validity Trade-off under Top-k Sampling

We evaluated the generative behavior of SeqGPT independently of the deterministic beam-search solver by sampling 25 sequences for each of 200 buckling targets. Diversity was controlled through *Top-k* sampling, with $k = 1$ corresponding to greedy decoding and $k > 1$ introducing stochastic exploration. No generated sequence appeared in the training set, indicating that the model learns physical correlations rather than memorizing configurations.
Figure 6 shows that while the model captures most industrial rules, increasing k degrades validity. Given the small vocabulary size ($|V| = 4$), setting $k = 4$ effectively removes any pruning of the output distribution. As a result, the sampler is forced to consider all possible ply orientations, including low-probability but physically invalid transitions (e.g., $0° \rightarrow 90°$), which leads to occasional violations of the disorientation rule.
A moderate setting ($k = 2$) provides a trade-off between diversity and feasibility, but strict manufacturing compliance without external repair requires greedy decoding ($k = 1$).

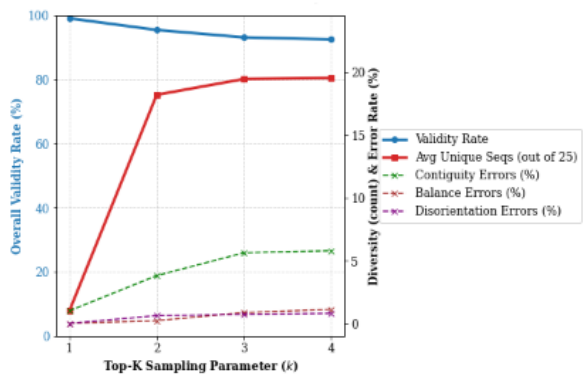

Figure 6: Evolution of the sequence validity rate (left axis) and the breakdown of specific constraint violations (right axis) as a function of the top-K sampling parameter.

4.1.2 Accuracy of Mechanical Target Conditioning

An effective inverse design framework must accurately recover the continuous physical properties prescribed by the macroscopic optimization phase. We evaluated the constrained model capacity to target specific parameters $(\alpha, \beta, \gamma, \delta)$ on 1000 unseen target vectors using greedy decoding ($k = 1$).
Exact matching of the continuous targets is often impossible due to the discrete nature of ply angles ($\{0°, \pm 45°, 90°\}$) and the cubic dependence of flexural stiffness. Consequently, targeting performance is evaluated

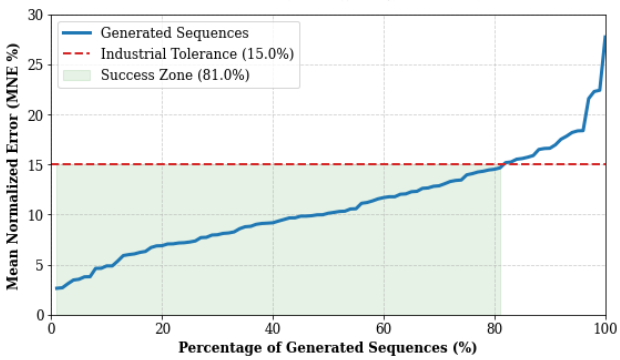

Figure 7: Cumulative distribution of the Mean Normalized Error (MNE) across 1000 generated sequences. The dashed line defines the 15.0% industrial tolerance threshold.

using the Mean Normalized Error (MNE). Figure 7 displays the cumulative distribution of the targeting error. Using a 15% MNE industrial tolerance threshold, the framework achieves an overall success rate of **81.0%**.

These controlled error rates confirm that the latent conditioning mechanism guides the generative process toward the required mechanical regions. SeqGPT then acts as an efficient heuristic locator that guides the beam search presented in the previous section.

4.1.3 Model Interpretability: Self-Attention Analysis

To understand the model's decision-making process, we analyzed the multi-head self-attention weights from the final transformer layer. These weights represent the network's focus on previously generated plies and conditioning targets during autoregressive decoding.

The visualizations in Fig. 8 suggest that the attention heads might capture different aspects of the sequence. As expected from causal masking, all matrices exhibit a lower-triangular structure. Head 2 (Fig. 8b) shows a strong concentration of attention weights along the immediate sub-diagonal. This suggests a primary focus on the directly preceding ply, which is consistent with the model learning local Markovian constraints (such as contiguity or maximum disorientation). In contrast, Head 1 (Fig. 8a) captures longer-range dependencies across the entire sequence history, which could indicate an aggregation of global context, potentially to monitor cumulative properties like the overall $\pm 45°$ balance.

Physical Alignment. The first column of both matrices shows how the model attends to the initial conditioning token ([PHYS]). Head 1 maintains persistent attention on this token, ensuring continuous grounding in the physical targets. Conversely, Head 2's attention to [PHYS] progressively decays as the sequence lengthens. Towards the end of the generation, local ply-to-ply constraints (the sub-diagonal) become so dominant that the model focuses almost exclusively on the preceding ply and ignores both the distant sequence history and the initial physical token. The third attention head is omitted due to redundancy.

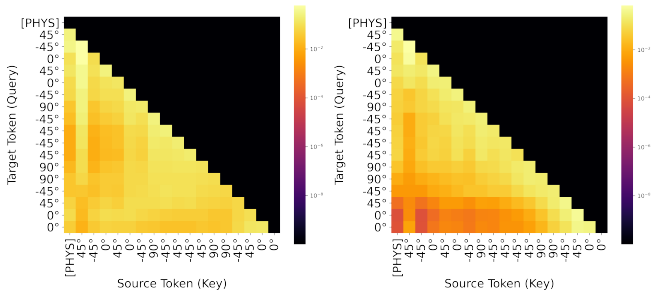

Figure 8: Self-attention weight distributions from the final layer of the SeqGPT model. (a) Head 1: Global context. (b) Head 2: Local Markovian constraints.

4.2 Multi-Panel Generation: 18-Panel Horseshoe Benchmark

4.2.1 Test Case

The performance of the proposed neuro-symbolic approach is evaluated on the reference 18-panel horseshoe benchmark ($r = 18$) [2]. As shown in Figure 9, this problem consists of interconnected panels with fixed biaxial loadings and dimensions. The structural material is Graphite/Epoxy IM7/8552 ($E_1 = 141$ GPa, $E_2 = 9.03$ GPa, $G_{12} = 4.27$ GPa, $\nu_{12} = 0.32$, ply thickness 0.191 mm). Panel thickness is bounded between $n_{min} = 14$ and $n_{max} = 48$ plies. While the original benchmark by Irisarri et al. [2] explores an extended orientation set (steps of 15°), we restrict our design space to the industrial standard set $\{0°, \pm 45°, 90°\}$. This choice aligns with conventional manufacturing certifications and focuses the challenge on achieving the target nemeth parameters with a more constrained, discrete vocabulary. The objective is to minimize total mass while ensuring a buckling Reserve Factor $RF > 1$. We retrieve optimal thickness distributions and buckling parameters directly from [2] to use as physical targets for sequence generation.

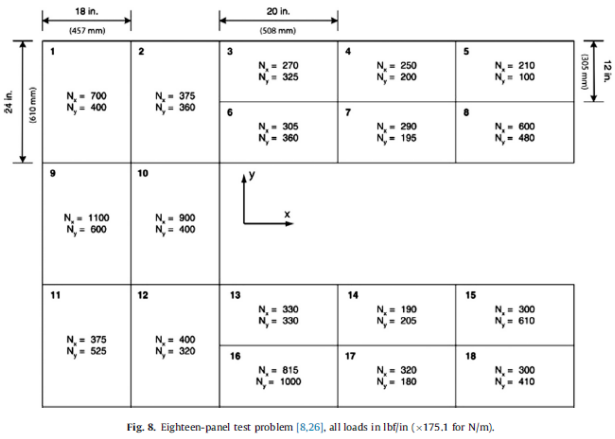

Figure 9: Eighteen-panel test problem benchmark layout [2]. All local loadings N_x and N_y are in lbf/in.

4.2.2 Propagation Map Definition

To translate the local geometries and biaxial loadings defined in the test case (Fig. 9) into manufacturable laminates, the multi-panel optimization process follows the standard bi-level strategy introduced in Section 2.1.

The first step of this framework—the continuous optimization phase—takes these geometric boundaries and internal loads (N_x, N_y) as inputs to minimize the total structural mass. This continuous relaxation yields the optimal target vector $\boldsymbol{\xi}^*$ alongside the target thickness N_{target} for each individual panel i. This output can be formally defined as the set of continuous physical targets $\mathcal{T} = \{(\boldsymbol{\xi}_i^*, N_{\text{target},i})\}_{i=1}^{18}$. Because the core focus of our study is the discrete stacking sequence recovery, we bypass executing the continuous FEA ourselves. Instead, we directly extract this global optimum target set $\mathcal{T}$ from the reference solution provided by Irisarri et al. [2].

To solve the subsequent multi-panel recovery problem—**the second step** of the bi-level optimization framework—we manually prescribe a Propagation Map using a Search Propagation Direction (SPD) strategy [6]. Figure 10 illustrates the topological layout for the 18-panel benchmark, where the integers denote the extracted target thicknesses $N_{\text{target},i}$. The map dictates the sequence generation order, propagating the design from designated start sequences to end sequences across adjacent panels. This definition effectively reduces the multi-panel blending problem into a sequential generation task, where topological continuity constraints are intrinsically handled by the neuro-symbolic beam search.

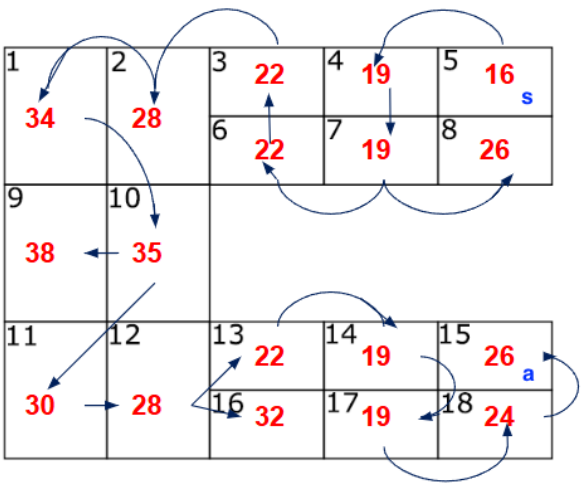

Figure 10: Propagation map for the 18-panel benchmark. Arrows indicate the Search Propagation Direction (SPD). The integers denote the extracted target thicknesses $N_{\text{target},i}$.

4.2.3 Comparative Analysis: SeqGPT vs. Evolutionary Algorithms

To assess the structural validity of the generated sequences, we compute the critical buckling load factor (λ_{cr}) for each panel. The resistance to buckling is governed by the panel's bending stiffness matrix $[D]$, which is directly derived from the stacking sequence and correlates with the lamination parameters $(\alpha, \beta, \gamma, \delta)$ targeted by the model.

For a simply supported rectangular plate of dimensions $a \times b$ subjected to biaxial compression (N_x, N_y), the buckling factor for a mode (m, n) is given by the analytical solution used in [2]:

$$\lambda_{(m,n)} = \frac{\pi^2\left[D_{11}\left(\frac{m}{a}\right)^4 + 2(D_{12}+2D_{66})\left(\frac{m}{a}\right)^2\left(\frac{n}{b}\right)^2 + D_{22}\left(\frac{n}{b}\right)^4\right]}{\left(\frac{m}{a}\right)^2 N_x + \left(\frac{n}{b}\right)^2 N_y}. \tag{4}$$

The Reserve Factor (RF) is defined as the minimum factor over all failure modes: $RF = \min_{m,n} \lambda_{(m,n)}$. A panel is considered safe if $RF > 1$. The values of a,b,N_x, and N_y are shown in Figure 9.

In this study, stability is reported using the Buckling Margin:

$$\text{Margin } (\%) = (RF - 1) \times 100 \tag{5}$$

Generated sequences are compared against "Solution 0" from Irisarri et al. [2], which serves as the reference for target thicknesses (N_{target}) and buckling margins.

Table 3 details the comparative results. Notably, for all 18 panels, SeqGPT generated fully blended stacking sequences matching the exact target thickness $(N_{\text{generated}} = N_{\text{target}})$. Because no sequence required an artificial thickness increment to satisfy the blending rules, **the conservative fallback strategy was never triggered**. This demonstrates the efficiency of the neuro-symbolic beam search in navigating a highly constrained design space on its first attempt, yielding structural buckling margins that are highly competitive with the reference Evolutionary Algorithm (EA), while avoiding any mass penalty.

Table 3: Comparison of SeqGPT generated solutions vs. Evolutionary Algorithm (EA) (Irisarri et al. [2]).

Panel	Target	SeqGPT (Proposed)		EA Reference
	N_{target}	$N_{\text{generated}}$	Margin (%)	Margin (%)
1	34	34	16.1	16.6
2	28	28	3.1	3.7
3	22	22	13.4	14.5
4	19	19	6.6	7.7
5	16	16	5.3	6.5
6	22	22	1.9	2.9
7	19	19	3.2	4.3
8	26	26	8.5	9.6
9	38	38	3.4	4.8
10	35	35	3.6	4.5
11	30	30	10.4	10.3
12	28	28	4.3	3.3
13	22	22	2.7	7.8
14	19	19	9.4	14.3
15	26	26	6.2	6.2
16	32	32	8.5	7.9
17	19	19	1.3	5.8
18	24	24	16.2	20.5

4.3 Discussion: Design Space, Performance, and Computational Cost

Comparing our generated sequences with the EA reference [2] highlights three key methodological differences:

1. **Restricted Design Space:** The reference EA operates on a wide range of orientations ($\{0°, \pm 15°, \ldots, 90°\}$). In contrast, SeqGPT is strictly restricted to standard aerospace angles $\{0°, \pm 45°, 90°\}$ to comply with industrial manufacturing standards. Despite this significant reduction in the design space, our approach yields competitive results that remain close to the baseline.

2. **Computational Efficiency:** The stochastic nature of EAs typically requires thousands of evaluations (approx. one hour for the reference study [2]). The proposed constructive approach dynamically prunes invalid sequences during the beam search, generating the entire set of blended sequences in a single inference pass. This reduces the total recovery time to under **4.5 seconds** (with a beam width of 10).

3. **Material Independence and Generalization:** The model was trained on a synthetic dataset of 20k sequences generated via HPC-parallelized Monte Carlo sampling using IMA/M21E material properties ($E_1 = 154$ GPa). However, the evaluation was performed on the IM7/8552 benchmark ($E_1 = 141$ GPa). The successful deployment on this new material system without retraining illustrates that SeqGPT does not overfit to specific stiffness values but learns a universal geometric mapping based on Nemeth parameters. Consequently, the proposed approach is inherently transferable to any other composite material system.

Conclusion

This paper introduced SeqGPT, a neuro-symbolic framework for the inverse design of blended composite structures under manufacturing constraints. By reformulating stacking sequence retrieval as a conditional sequence generation problem and combining a lightweight Transformer with constrained decoding, the approach eliminates the main computational bottleneck of classical bi-step composite optimization workflows.

The key contribution lies in the explicit decoupling between data-driven modeling of mechanical correlations and symbolic enforcement of blending rules, ensuring manufacturable solutions by construction. Results on the 18-panel horseshoe benchmark demonstrate buckling performances comparable to state-of-the-art evolutionary approaches, with recovery times reduced from hours to seconds.

Several directions for future work are identified to address current limitations and extend the framework. First, the current formulation aggregates all physical parameters into a single context embedding. Future developments will investigate more structured conditioning schemes based on multiple context tokens, enabling finer control of the self-attention mechanism and improving the alignment between continuous performance targets and discrete stacking sequence realizability. Second, multi-panel composite structures are naturally represented as graphs. While the current Search Propagation Direction (SPD) is manually defined through template-based strategies, future work will explore systematic graph-based formulations of propagation rules. Such an approach would enable the automatic inference of propagation paths for complex topologies while preserving the guarantees provided by symbolic constrained decoding.

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Generative AI Usage

In accordance with the European Commission's recommendations, the authors acknowledge the use of generative AI (Google Gemini) for English language revision and assistance with TikZ figure generation. The authors assume full responsibility for all scientific content, methodologies, and conclusions presented in this manuscript.

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